Nutan Maharashtra Vidya Prasarak Mandal's

NUTAN MAHARASHTRA INSTITUTE OF ENGINEERING & TECHNOLOGY, PUNE

Department of Information Technology

NAAC & NBA Accredited



SPPU Project Work Stage – I

Academic Year 2026-27

Nutan Maharashtra Vidya Prasarak Mandal's
NUTAN MAHARASHTRA INSTITUTE OF ENGINEERING & TECHNOLOGY, PUNE
Department of Information Technology

A
Project Review
on

"A RAG-Based Multilingual AI Assistant for Government Scheme Discovery and Application Assistance"

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- **Content:**
- **Project Presentation**
  - Project Title,
  - Originality of Problem Statement,
  - Depth of Understanding the Problem Statement,
  - Concreate Literature Survey with identified gaps in all referred papers,
  - Design and Analysis of Algorithm / Model / Architecture / System,
  - List of required hardware, software, or other equipment for executing the project,
  - project Scope & objective,
  - System overview,
  - Architecture and initial phase of design (DFD),
  - Expected outcome

- **Project Title**

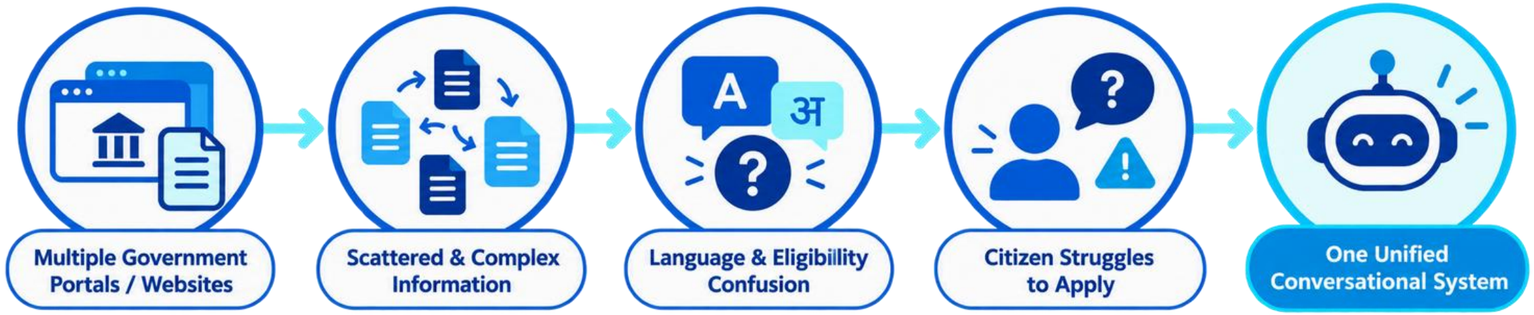
A RAG-Based Multilingual AI Assistant for Government Scheme Discovery and Application Assistance

- **Problem Domain**

RAG (Retrieval-Augmented Generation, Artificial Intelligence & Machine Learning (AI/ML)

- **Originality of Problem Statement**

Citizens in India, especially in semi-urban and rural areas, struggle to access accurate, up-to-date information about government schemes, certificates, and eligibility criteria. Information is scattered across multiple government portals, often in complex bureaucratic language, primarily in English or formal Marathi/Hindi, making it inaccessible to citizens with limited literacy or digital skills. There is no unified, conversational system that helps citizens discover relevant schemes, check eligibility, and prepare applications with ease.



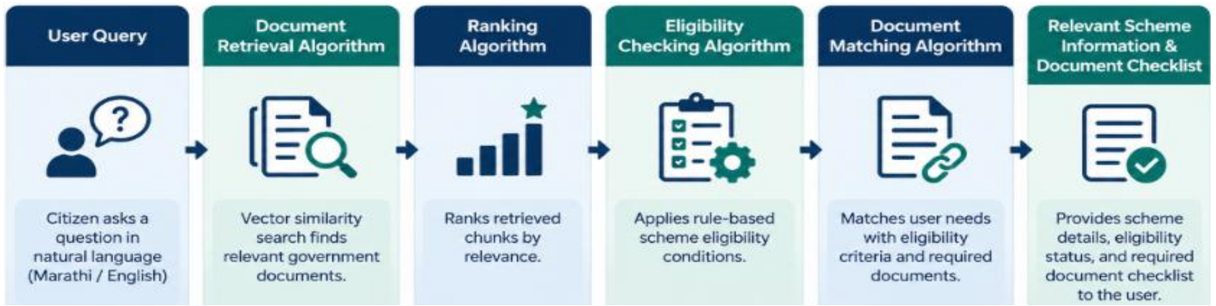
- **Depth of Understanding the Problem Statement**

Citizens often do not know which government schemes or certificates they are eligible for, and even when they do, the application process is confusing and paperwork-heavy. Existing government websites are static, text-heavy, and not conversational, offering no personalized guidance.
GovKnowledge Hub addresses this by providing a chatbot-based AI assistant, accessible via website and WhatsApp, that ingests government scheme data (eligibility rules, documents required, procedures) into a Retrieval-Augmented Generation (RAG) pipeline. Users can converse in natural language — including Marathi — and ask questions like "Which scholarships can I apply for?" or "Am I eligible for X certificate?" The assistant retrieves relevant scheme information from a vector database, grounds its answers with source citations to minimize hallucination, and additionally helps the user by auto-generating a pre-filled application draft/summary (name, details, required documents checklist) that the user can review and submit on the actual government portal.

- **Concreate Literature Survey with identified gaps in all referred papers,**

Sr.	Year	Paper / Study (Authors)	Key Findings	Methods	Limitations	Link
1	2026	**Government Schemes Recommendation (Sheriff et al.)**	RAG improves access to government scheme information.	FAISS, LangChain, GenAI, React, Node.js, FastAPI	Limited schemes; English/Tamil only.	https://www.ijfmr.com/papers/2026/3/79624.pdf
2	2026	**AI-Driven Chatbot for Govt Scheme Accessibility (Vaishnavi et al.)**	Hybrid RAG achieves 90.4% retrieval accuracy.	BM25, Dense Embeddings, Reranking, Local LLM	Requires updated scheme database.	https://ijirt.org/publishedpaper/IJIRT201405_PAPER.pdf
3	2026	**Groundedness in Government Document Chatbots (Rumamby et al.)**	Identifies weaknesses in existing RAG evaluation metrics.	Systematic Literature Review, PRISMA	Proposed metric needs validation.	https://doi.org/10.70062/slrj.v2i1.277
4	2026	**AgriGov: Structured Multilingual Dataset (Bilal & Gopakumar)**	Provides multilingual agricultural scheme data.	Web Scraping, Machine Translation, Human Review	Focuses only on agricultural schemes.	https://arxiv.org/abs/2606.08272
5	2025	**Government Schemes Recommendation API (Reddy et al.)**	Multilingual RAG simplifies welfare scheme discovery.	FAISS, LangChain, FastAPI, Eligibility Filtering	Limited schemes and languages.	https://doi.org/10.1109/ICIMIA67127.2025.11200536
6	2024	**Need of RAG for Conversational Assistants (Vaidheeswaran & Nathan)**	RAG improves answer quality and reduces hallucinations.	Comparative Experiments, Hybrid Retrieval, ASR/TTS	Limited to the studied welfare-scheme domain.	https://doi.org/10.30574/ijsra.2024.13.1.1747
7	2024	**Two-Phase RAG Chatbot for Italian Funding (Boccato et al.)**	Two-stage RAG improves information retrieval.	Two-Phase RAG, Summary-Based Retrieval	Domain-specific; moderate user satisfaction.	https://www.preprints.org/manuscript/202406.1999

- **Design and Analysis of Algorithms**
  - **Algorithms Used:**
    - Document Retrieval Algorithm: Retrieves relevant government documents based on user queries using vector similarity search.
    - Ranking Algorithm: Ranks retrieved document chunks according to their relevance to the user's question.
    - Eligibility Checking Algorithm: Applies rule-based conditions to check users' eligibility for government schemes.
    - Document Matching Algorithm: Matches user requirements with scheme eligibility criteria and required documents.



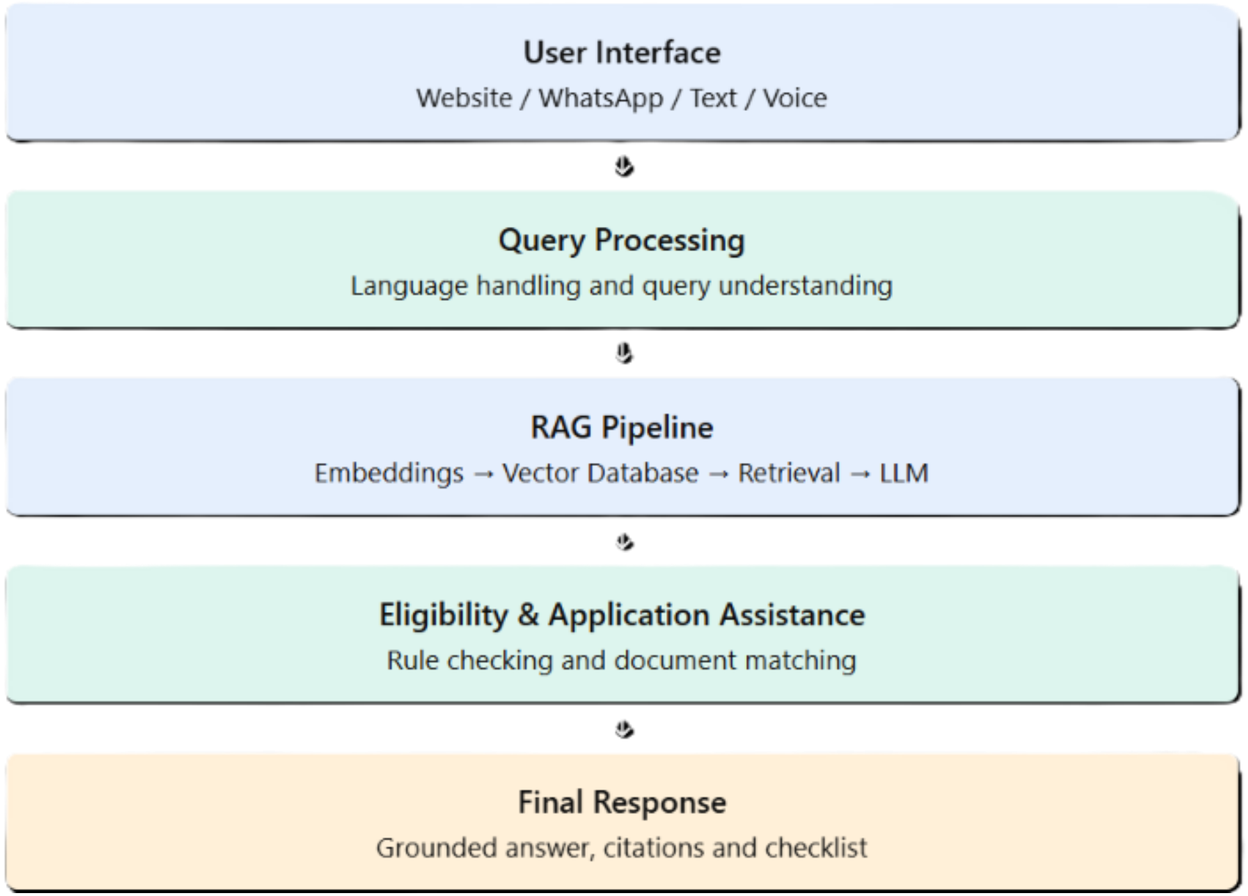
- **Algorithm Analysis:**

- Time Complexity: Depends on the retrieval method, dataset size, and number of eligibility rules.
- Space Complexity: Depends on the number of stored document chunks, embeddings, and indexing structures.
- Optimization: Efficient indexing and retrieval reduce search time and improve response efficiency.

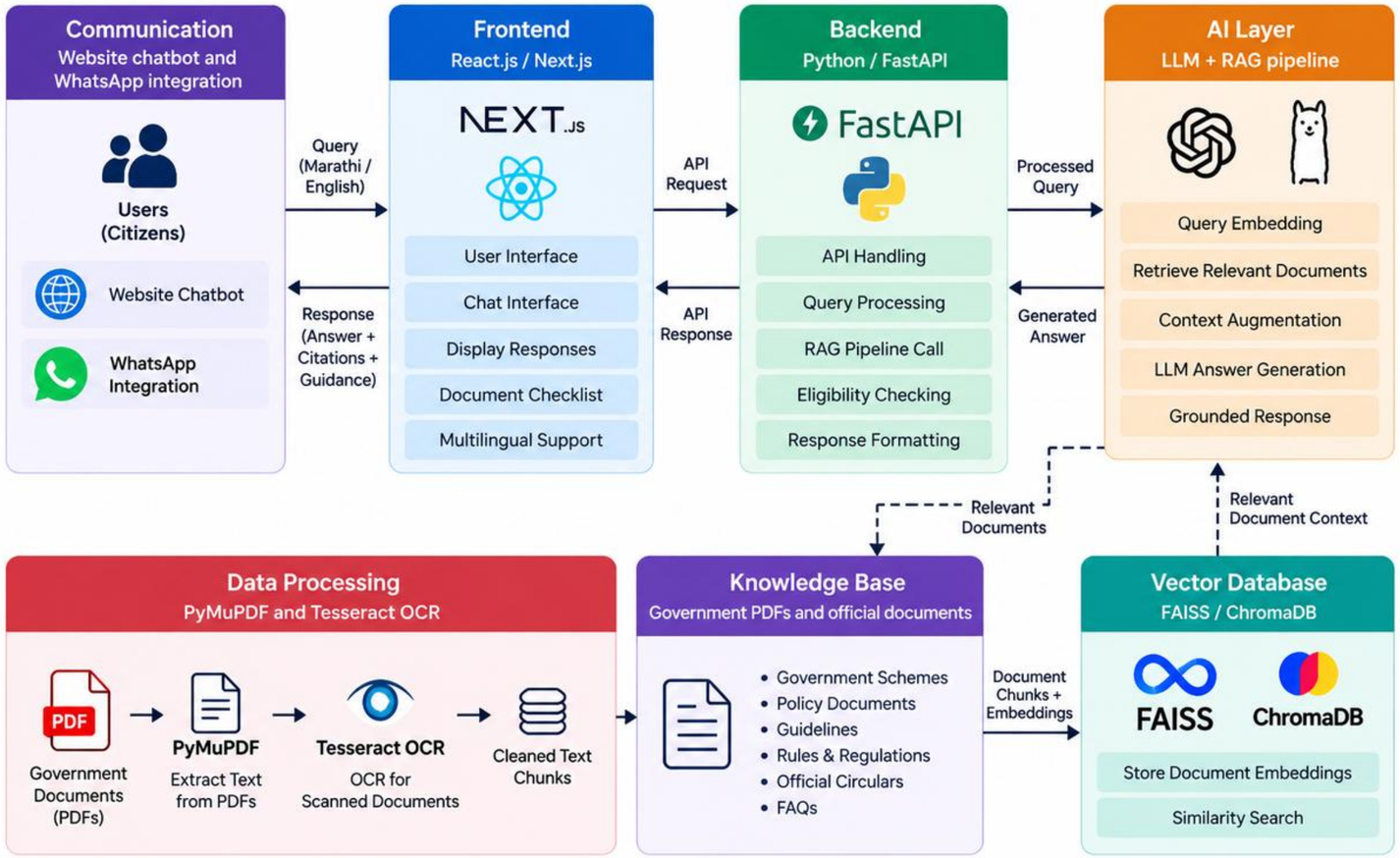
- **Design of Model**

- Input: User query in Marathi or English.
- Embedding Model: Converts user queries and government documents into vector representations.
- RAG Model: Retrieves relevant document context and generates answers using an LLM.
- Output: Provides answers with source citations, eligibility guidance, and a required document checklist.

- **System Architecture:**



- **System Design**



- **List of Required Hardware, Software, and Other Equipment**

1. Hardware Requirements

- Processor: Intel Core i5 / AMD Ryzen 5 or equivalent
- RAM: Minimum 8 GB (16 GB recommended)
- Storage: Minimum 256 GB SSD
- Internet Connection: Stable broadband connection
- Display: Standard monitor or laptop display

2. Software Requirements

- Operating System: Windows 10/11 or Linux
- Frontend: React.js / Next.js, HTML, CSS, JavaScript
- Backend: Python and FastAPI
- AI / LLM: OpenAI API or a compatible LLM
- RAG Framework: LangChain
- Vector Database: FAISS or ChromaDB
- Document Processing: PyMuPDF and Tesseract OCR
- Development Tools: Visual Studio Code, Git, GitHub, Postman
- Database: SQLite or PostgreSQL, if required
- Browser: Google Chrome or Microsoft Edge

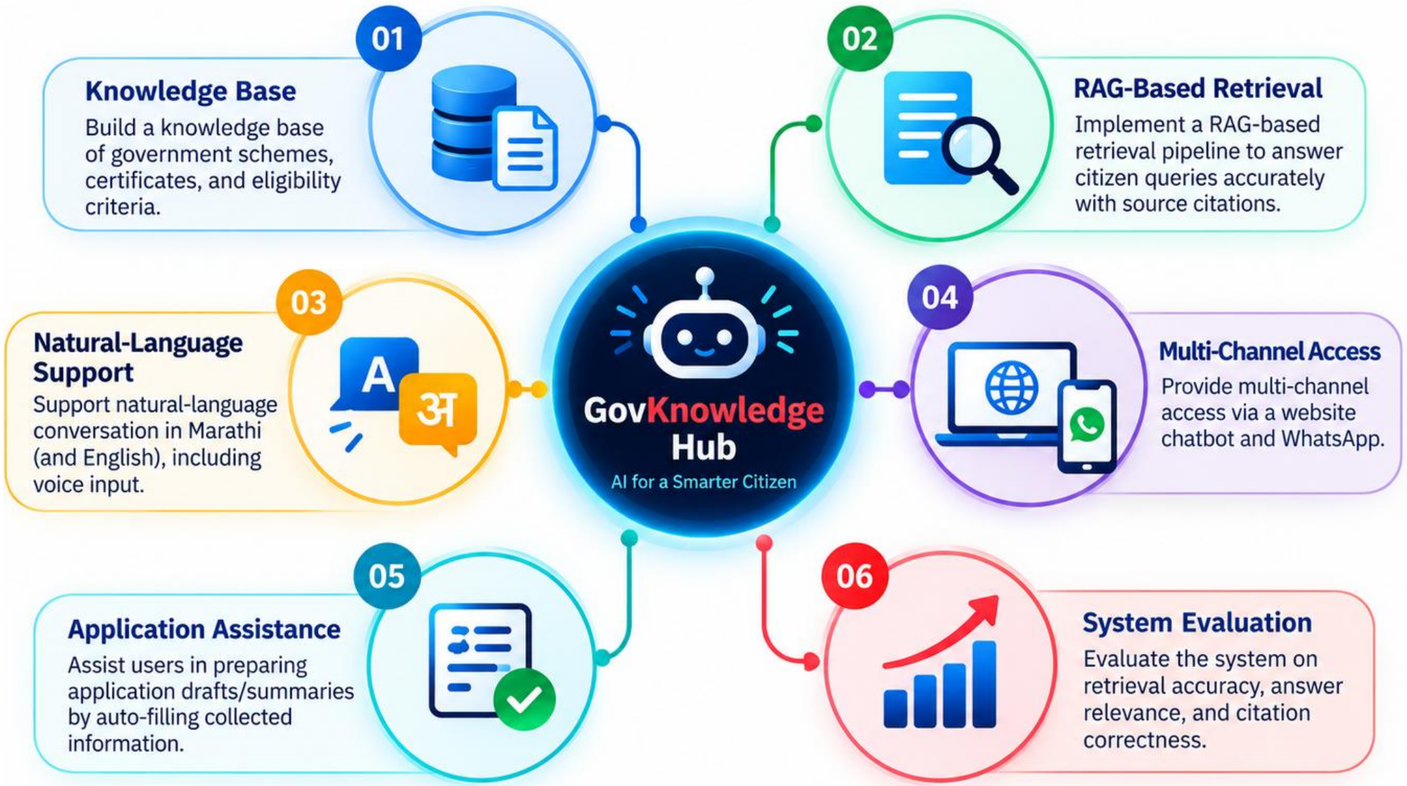
3. Other Requirements

- Official government PDFs, scheme guidelines, GRs, and circulars
- API key and internet access for the selected LLM service
- Government document datasets for testing and validation
- WhatsApp Business API access, if WhatsApp integration is implemented
- Cloud hosting or a deployment platform, if online deployment is required

- **Project Scope**

- Develop an AI-powered platform for accessing government schemes, rules, GRs, and official documents.
- Implement a Retrieval-Augmented Generation (RAG) system to provide answers based on relevant government sources.
- Support multilingual interaction in Marathi and English.
- Provide source citations to help users verify the information.
- Assist citizens in understanding scheme eligibility criteria and required documents.
- Generate document checklists and application guidance.
- Provide a user-friendly web interface with optional WhatsApp and voice integration.
- Evaluate retrieval accuracy, answer relevance, citation correctness, and response reliability.

- **Project Objectives**



- **System Overview**

GovKnowledge Hub is an AI-powered platform designed to simplify access to government schemes, rules, GRs, and official documents. It uses Retrieval-Augmented Generation (RAG) to retrieve relevant information from official government sources and generate context-based answers using a Large Language Model (LLM).
The system supports Marathi and English queries, provides source citations for verification, and assists citizens in understanding eligibility criteria, required documents, and application procedures.

- **System Workflow**

  - **User Input:** Citizens submit queries in Marathi or English through the web interface.
  - **Query Processing:** The system processes the query and prepares it for retrieval.
  - **Knowledge Retrieval:** Relevant information is retrieved from the government document knowledge base using embeddings and vector similarity search.
  - **Answer Generation:** The LLM generates a response based on the retrieved context.
  - **Eligibility Assistance:** The system helps users understand scheme eligibility and document requirements.
  - **Response Delivery:** The system presents the answer with source citations, eligibility guidance, and a document checklist.

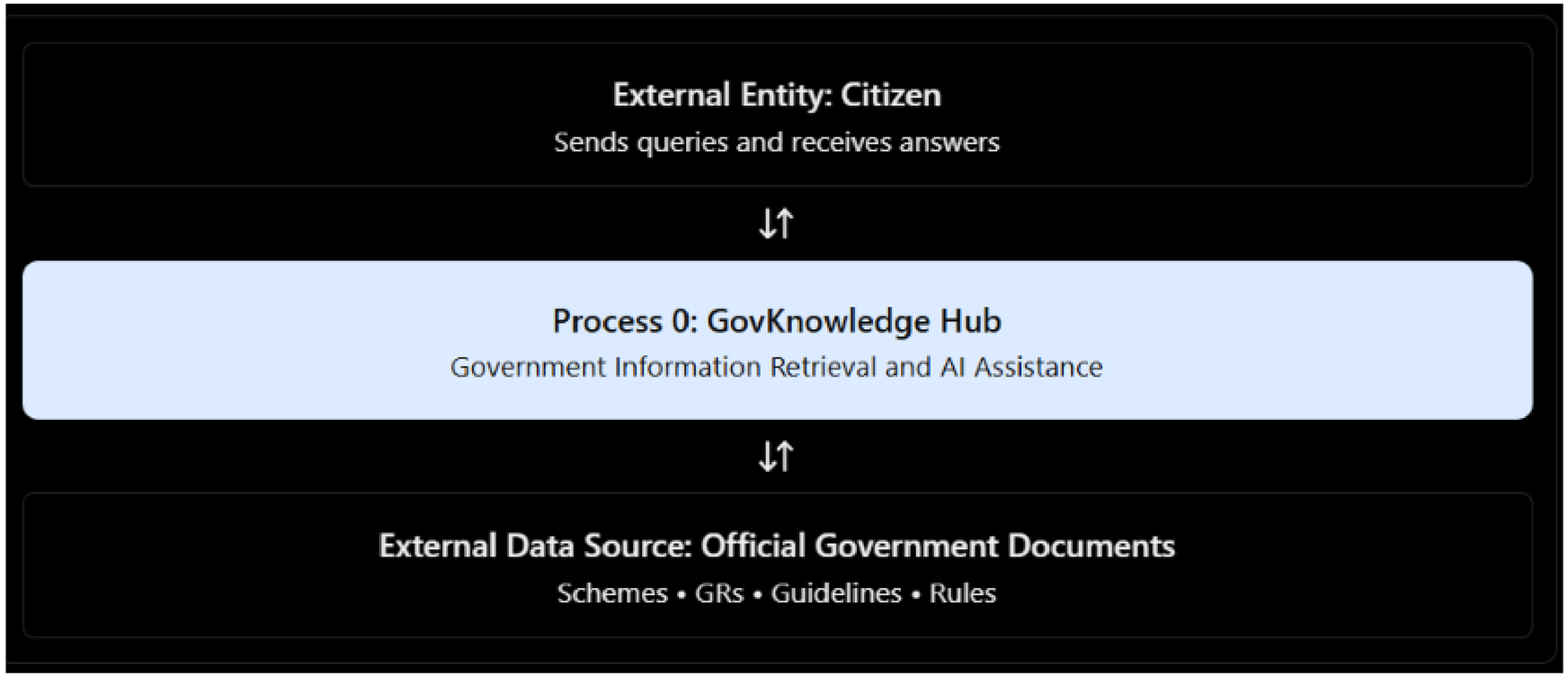
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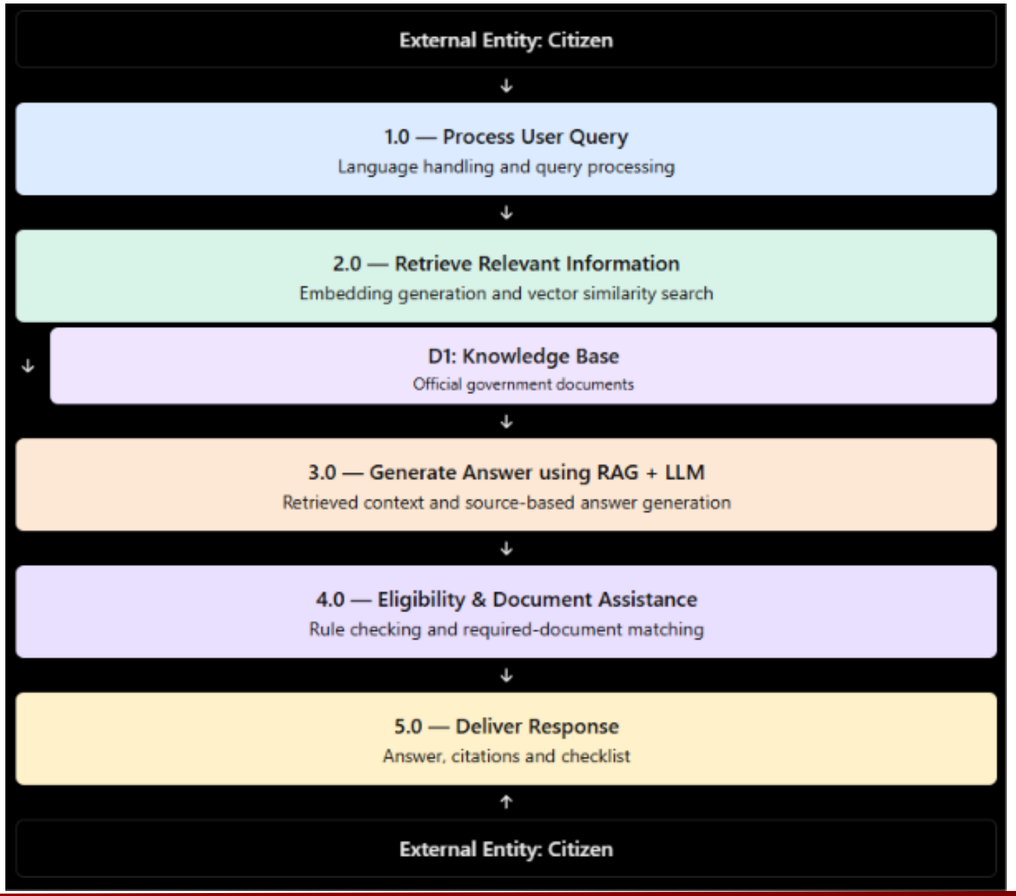
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- **Initial Phase of Design — Data Flow Diagram (DFD)**
  - **DFD Level 0 — Context Diagram**



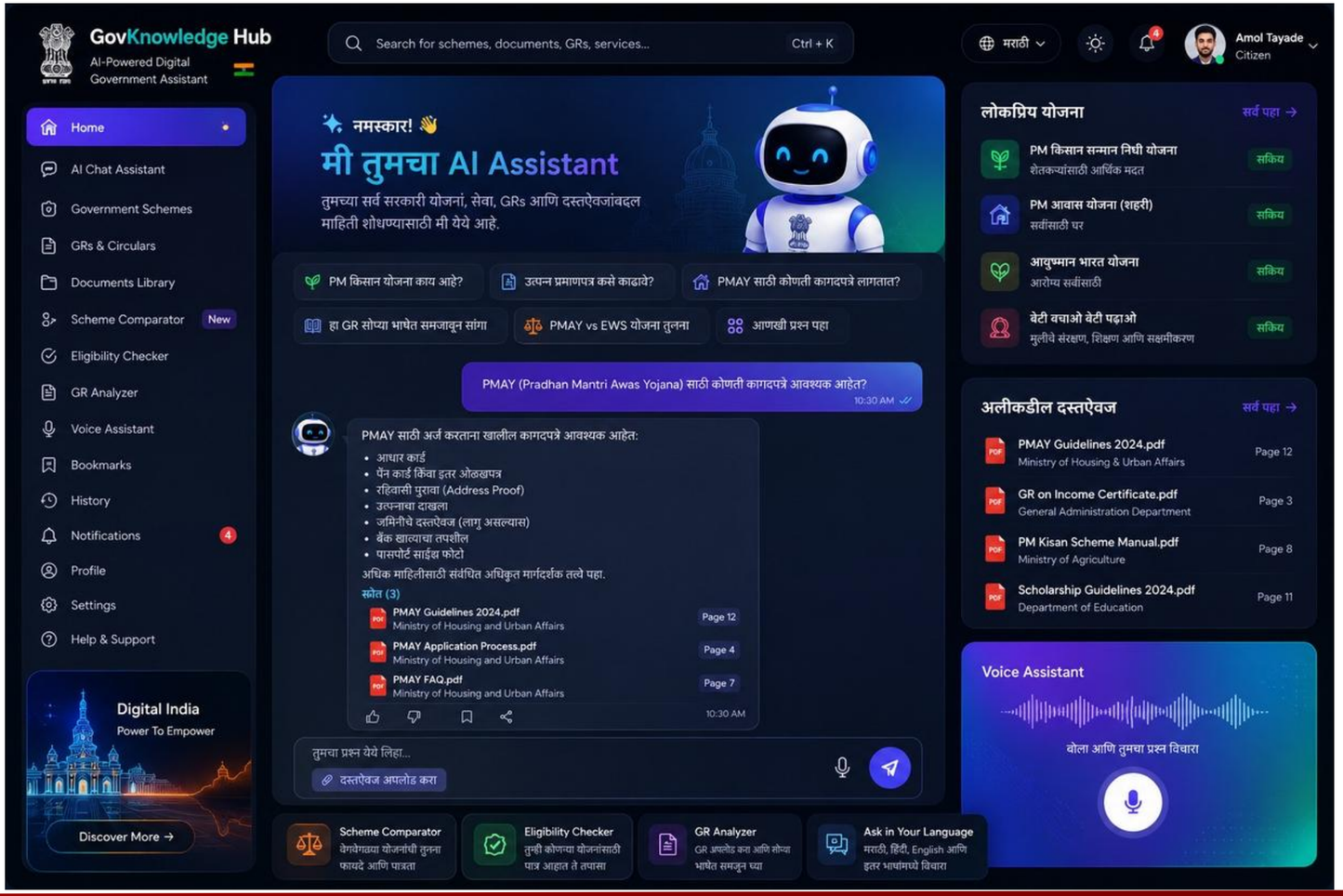
- **Initial Phase of Design — Data Flow Diagram (DFD)**
  - **DFD Level 1 — Detailed Data Flow**



Expected outcome

- Centralized Government Information: A unified platform to access government schemes, GRs, rules, guidelines, and official documents.
- Accurate AI Responses: RAG-based answers generated using relevant government document context.
- Multilingual Accessibility: Support for queries in Marathi and English.
- Source-Based Answers: Citations that help users verify information against official documents.
- Eligibility Assistance: Simplified guidance on government scheme eligibility criteria.
- Document Checklist Generation: Identification of required documents for schemes and government services.
- Improved User Experience: Easy access to government information through a user-friendly web interface.
- Reliable Information Retrieval: Evaluation of retrieval accuracy, answer relevance, citation correctness, and response groundedness.

- **Expected outcome**



Thank You

FOR YOUR ATTENTION